# PULSE: Data, Database & AI: Full Report

Everything the data/AI workstream built for the PULSE hypertension-monitoring prototype: how the parts relate, every table, every dataset, and every model with its training data, test data and results.

30 October 2026 · Data/AI workstream · Branch `AI` of `github.com/Ayoub-Meziani/Pulse` · Cloud project `pulse` (Supabase, Paris). Prototype, not a medical device. Patient data is **fictional** and labelled `simulated`. Real public datasets were used only for statistics, calibration and tests.

| | | | |
|---|---|---|---|
| **8 / 8**<br>migrations, cloud in sync | **24**<br>tables, all with access rules (52 policies) | **14**<br>demo accounts (1 doctor, 1 admin, 12 patients) | **730**<br>blood-pressure readings in the cloud |
| **20,832**<br>simulated ring samples (12 patients × 28 days) | **6**<br>models: 3 in use, live early warning, 1 experiment, 1 not shown | **3**<br>real public datasets (NHANES, Mendeley, PhysioNet) | **95**<br>automated tests (database, data, ML) |

**Status labels:** LIVE in the cloud database · BUILT code exists and is tested · EXPERIMENT evaluated, not used · TO BUILD another owner's next step · MISSING nobody has it yet · TO VERIFY check before use

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## 1. Where we are

| Part | Status | Details |
|---|---|---|
| Cloud database `pulse` | LIVE | Migrations M1–M9 applied (M9 kept by Ayoub's decision of 31 Oct). Cloud jobs and scripts frozen by Ayoub (incident report of 30 Oct). 24 tables, row-level security (RLS) on every table, 52 policies, nothing granted to logged-out users, public sign-up off. |
| Demo data in the cloud | LIVE | 14 accounts; 730 readings (727 valid, 2 needs_review, 1 invalid, on purpose); 316 lifestyle logs; 154 dose events; 11 care plans; 24 body measurements; 24 devices; 20,832 ring samples for all 12 patients. All labelled `simulated`. Password only in the local, git-ignored `.env`. |
| ML outputs in the cloud | LIVE | Jobs ran on the cloud on 30 Oct (`scripts\run_cloud_jobs.ps1`; without Ayoub's confirmed approval, see the incident report): 10 next-day BP estimates + 1 `insufficient_history`; 33 ring flags (22 outside reference, 10 personal, 1 multivariate); 11 weekly reports, all with the classifier probability; 43 proposed bands waiting for the doctor. |
| Alerts in the cloud | TO BUILD | 0 rows. The alert rules live in the backend (FastAPI, teammate), which still runs on SQLite. The tables and the episode logic (M3) are ready. |
| 6 models (`ml/`, Python) | BUILT | `mbp-v1.1` next-day BP (served), `ring-v1.1` ring flags, `wcls-v1.1` weekly classifier, `live-v1` live early warnings (section 6.10), `mbp-v2.1` (experiment), `wrisk-v1` on real NHANES people (evaluated, not shown). Section 6. |

| | | |
|---|---|---|
| Real public data | BUILT | NHANES 2017–2020 (CDC), Mendeley home-BP series, PhysioNet Apnea-ECG. Downloaded to git-ignored `data/external/` with a manifest, never committed or uploaded. Section 5.6. |
| Device (cuff / ring) | MISSING | No physical device yet. Tables for cuff readings and ring samples are live; ring data is simulated from published values. |
| Real patient data | MISSING | None, on purpose. No open multi-patient home-BP dataset exists, so real accuracy needs consented PULSE data later. |

## 2. System map: how everything connects

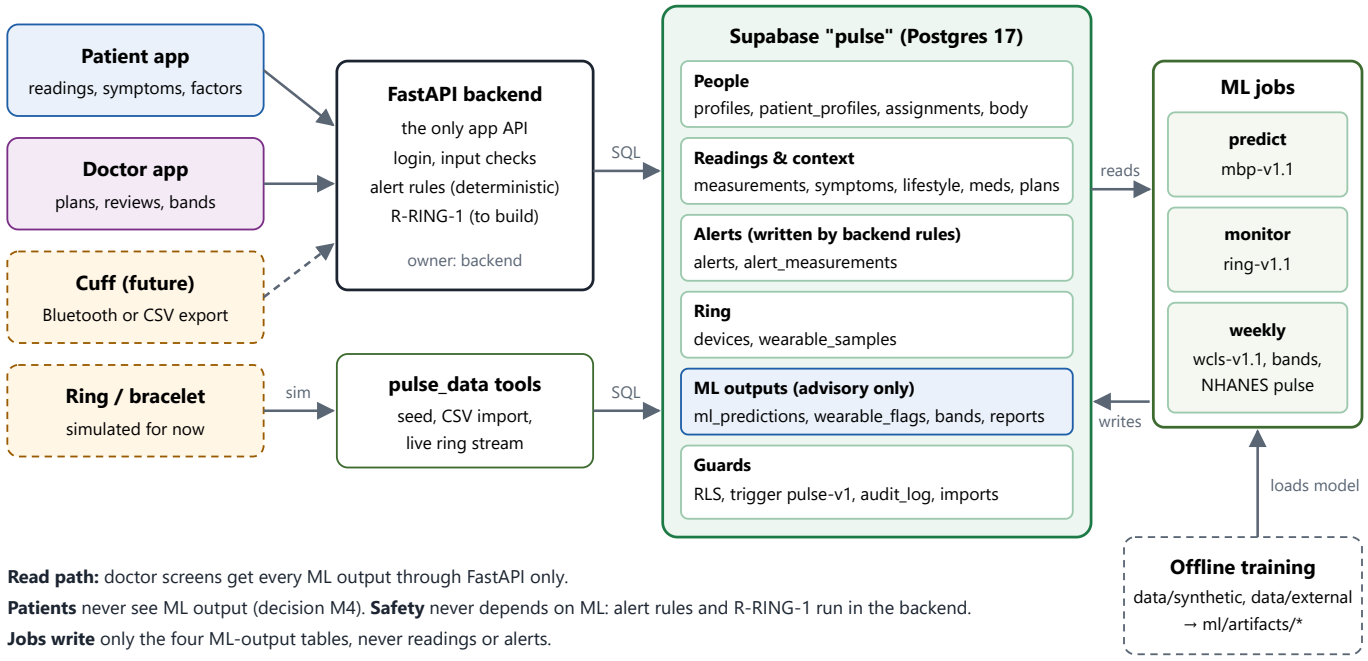



Figure 1. Data enters through FastAPI or the data tools, lands in Supabase, and the three ML jobs read it and write back only advisory outputs. Dashed = not built yet or simulated.

| Component | Talks to | Credential it uses | Where that credential lives |
|---|---|---|---|
| Frontend (patient, doctor) | FastAPI only (team decision A2) | the user's own login token | nothing stored |
| FastAPI backend | Postgres; Supabase Auth (JWT checked with JWKS, ES256) | database URL, Supabase URL, anon key | server `.env` / hosting secrets |
| ML jobs, CSV import, seed | Postgres directly | `DATABASE_URL` | data/AI machine `.env` (git-ignored) |
| Migrations | Supabase CLI `db push` | project owner's CLI login | never shared |

**Never:** the service-role key or the database password in the frontend, git, chat or screenshots. The GitHub repo is **public**.

## 3. What relates to what

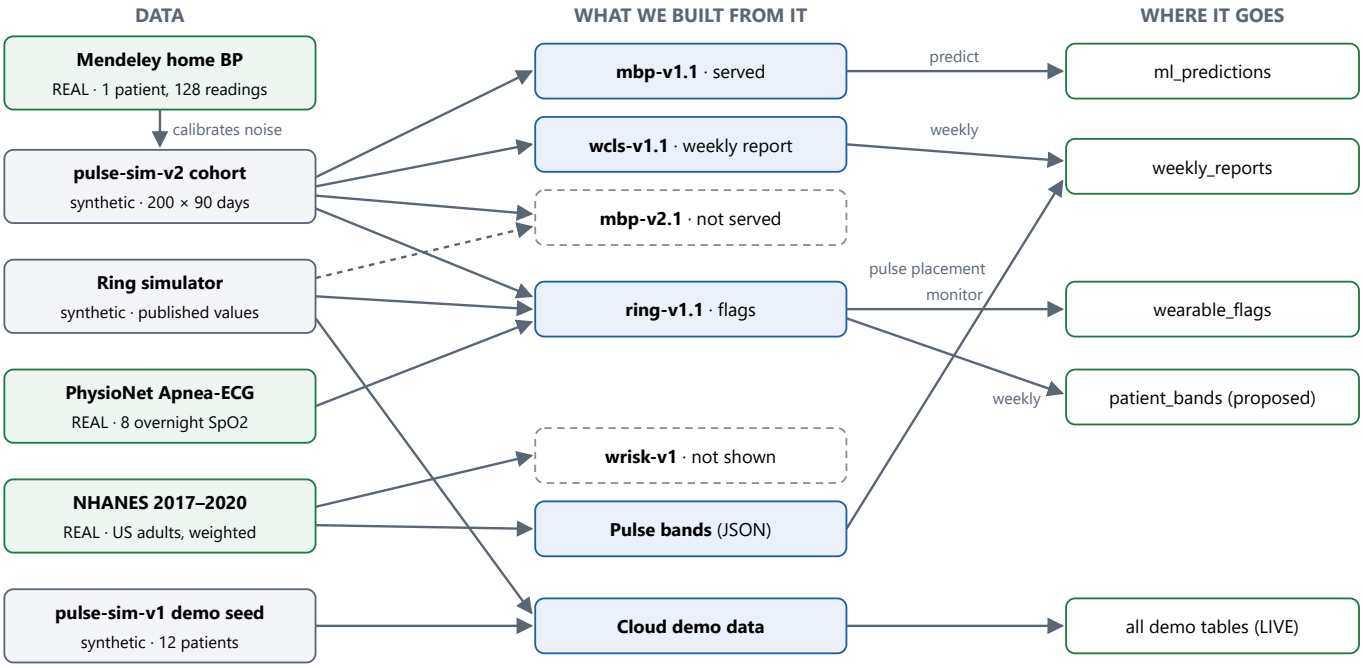



Figure 2. Every dataset, what it produced, and the table each result lands in. Mendeley calibrates the v2 simulator; Apnea-ECG tests the ring SpO2 rule; NHANES gives the pulse bands and wrisk-v1. Green = real public data; dashed = evaluated but not used in the product.

| Component | Code | Reads | Writes | Who sees it | Status |
|---|---|---|---|---|---|
| Validation trigger `pulse-v1` | migration M1 | each new reading | validation_status, quality_flags | everyone (on the reading) | LIVE |
| Access rules (RLS) | M2, M3, M6, M7 | caller's role + assignments | - | - | LIVE |
| Demo seed | `pulse_data/simulate.py seed` | STEPS-DZ shares, ring values | all demo tables | team, demo | LIVE |
| Cloud ring refresh | `simulate ring-demo-sql`, `scripts/cloud_update.ps1` | ring simulator | wearable_samples (simulated only) | doctor | LIVE |
| CSV import | `pulse_data/ingest.py` | device CSV export | measurements, ingestion_batches | patient, doctor | BUILT |
| Live ring stream | `pulse_data/stream.py` | ring simulator | wearable_samples | doctor (via flags) | BUILT (local) |
| Next-day BP | `ml/pulse_ml/predict.py` + `mbp-v1.1` | valid readings, assignments | ml_predictions | doctor only | BUILT |
| Ring monitor | `monitor.py`, `ring.py` + `ring-v1.1` | wearable_samples, active bands, conditions, confirmed flags | wearable_flags | doctor only | BUILT |
| Weekly report | `weekly.py` + `wcls-v1.1`, `bands.py`, NHANES bands | readings, care plans, ring nights, flags, predictions, profile | weekly_reports, patient_bands (proposed) | doctor only | BUILT |
| Band approval | `private.approve_band` (M7) | proposed band | patient_bands (active), audit_log | doctor acts | LIVE (API TO BUILD) |
| Feedback gate | `feedback.py` | reviewed flags | git-ignored labels file | data/AI | BUILT |
| Medical references | `reference.py` | - | - | used by monitor, weekly, bands | REVIEWED 30 Oct |

| | | | | | |
|---|---|---|---|---|---|
| Real-data tools | `real_data.py`, `real_bands.py`, `wrisk.py`, `real_tests.py` | data/external (git-ignored) | reference_nhanes.json, metadata | - | BUILT |
| Alert rules | backend `rules.py` | readings, symptoms, care plan | alerts, alert_measurements | patient + doctor | TO BUILD on Supabase |

# 4. Database

## 4.1 Migrations (one append-only history in `supabase/migrations/`)

| # | File | What it added, and why | Cloud |
|---|---|---|---|
| M1 | `20261029120000_core_schema` | 17 tables; validation trigger `pulse-v1`; audit triggers; role checks; profile row on sign-up. The base of every feature. | LIVE |
| M2 | `20261029120100_access_policies` | RLS on every table; `anon` gets nothing; per-role write policies; column grants (e.g. a patient cannot change their role). | LIVE |
| M3 | `20261030120000_backend_alignment` | Backend contract: `above_target_trend` alert class, `alerts.persistent`, table `alert_measurements` (same-patient keys), `measurements.rested_5_min`, empty symptom answers allowed, doctor notes hidden from patients. | LIVE |
| M4 | `20261030120100_ml_predictions` | Table `ml_predictions`: values present only when status is `ok`, low ≤ high; doctor/admin read only; no client writes. | LIVE |
| M5 | `20261030130000_patient_birth_year` | `patient_profiles.birth_year` (year only, data minimisation). Age is the strongest Algerian hypertension factor. | LIVE |
| M6 | `20261030140000_algerian_factors_wearables` | Sex, smoking, condition codes, bread / salt habits, language, fasting logs; tables `body_measurements` and `wearable_samples` (unit and plausible range checked per metric). | LIVE |
| M7 | `20261030150000_ml_monitoring` | Tables `wearable_flags`, `patient_bands`, `weekly_reports`; function `private.approve_band`; audit on all three; index for the live monitor. Built exactly as the backend's ML review asked. | LIVE |
| M8 | `20261030160000_chronic_conditions` | Condition codes add copd, asthma, obstructive_sleep_apnea, atrial_fibrillation, heart_failure, thyroid_disease (they change how ring values are judged). Applied 30 Oct, without Ayoub's confirmed approval (incident report). | LIVE |
| M9 | `20261030170000_live_early_warning` | Ring flags can be `early_warning` / `forecast` (and `sleep`) for the live engine `live-v1`: warn the doctor before a limit is crossed. Additive only, no row changed. In the cloud since 30 Oct (21:47–22:58 UTC); Ayoub decided on 31 Oct to keep it (not approved retroactively). `live-v1` stays inactive. | LIVE |

## 4.2 How the tables relate

- **One person = one login** (`auth.users`) = one `profiles` row (role patient / doctor / admin). Patients have one `patient_profiles` row.

- **Every patient-owned table** has `patient_id → profiles.id`. A doctor's access is never stored per row: it is computed from an **active** `doctor_patient_assignments` row by the helpers `private.is_assigned_doctor()` and `private.is_admin()`.
- **Same-patient keys:** a reading can only use the patient's own device; an alert can only attach the same patient's readings (`alert_measurements`, composite keys); ring samples can only come from the patient's own device.
- **Chains:** `measurements → alerts → alert_measurements`; `devices → wearable_samples → (monitor) → wearable_flags → band_id → patient_bands`; `patient_bands.supersedes_id` and `care_plans.supersedes_id` keep history instead of overwriting.
- **History, never silent deletes:** readings are append-only; wrong readings stay, flagged; changes to sensitive tables are copied to `audit_log` (old row, new row, actor).

## 4.3 Every table and field (24 tables)

Every table also has `id` (except link tables) and `created_at`, filled automatically. **Blue** = patient, **purple** = doctor, **green** = system / device / admin. Column counts are from the cloud.

### People and access

| Table | Purpose | Patient fills | Doctor fills | System / admin |
|---|---|---|---|---|
| **profiles (7)** | One row per account | display_name, timezone (default Africa/Algiers), preferred_language (fr / ar) | same | id = login id, role (admin only), updated_at |
| **patient_profiles (18)** | Background and risk factors | birth_year, sex, smoking_status, condition_codes, relevant_conditions, bread_portions_per_day, adds_salt_at_table, salty_processed_food, allergies, medication_context, regular_clinic, emergency_preferences, consent_location, data_consent_at | (reads) | user_id, regular_doctor_id, updated_at |
| **body_measurements (9)** | Weight, height, waist | measured_at, weight_kg, height_cm, waist_cm (patient_reported) | same (clinician) | created_by; no edits or deletes |
| **doctor_patient_assignments (7)** | Who follows whom; drives all doctor access | (reads) | (reads) | doctor_id, patient_id, status active / ended, ended_at |
| **audit_log (8)** | Who changed what | - | - | table_name, row_id, action, actor, changed_at, old_data, new_data (admin reads) |

### Readings, devices, ring

| Table | Purpose | Patient fills | Device / import | System |
|---|---|---|---|---|
| **measurements (24)** | Every BP reading (append-only) | systolic, diastolic, pulse, measured_at, measured_tz, body_position, arm, context, rested_5_min | device_id, ingestion_batch_id, external_id, method, raw_payload, device flags | source, provenance, unit mmHg, validation_status, quality_flags, validation_policy_version, created_by |
| **devices (9)** | Cuffs, rings, phone apps | device_type, manufacturer, model, external_device_id, label, bp_capability | - | patient_id |

| | | | | |
|---|---|---|---|---|
| **wearable_samples (10)** | Ring / bracelet values | - | device_id, metric (heart_rate bpm, hrv_rmssd ms, spo2 percent, skin_temperature celsius), value, measured_at, external_id | source (device_sync / csv_import / simulated); unit + range checked; one row per (device, external_id) |
| **ingestion_batches (13)** | One row per import or sync | - | source, device_id, file_name, file_sha256 (same file never twice) | row counts, rejections (row + reason, no health values) |

### Patient-reported context

| Table | Purpose | Patient fills | System |
|---|---|---|---|
| **symptom_reports (8)** | Symptoms, judged independently of BP | symptoms (none, or chest_pain, shortness_of_breath, back_pain, weakness, numbness, vision_change, difficulty_speaking, severe_headache, confusion, dizziness, nausea, palpitations, other), onset_at, notes | reported_at, source |
| **lifestyle_logs (17)** | Meals, activity, sleep, stress, fasting | category, started_at, ended_at, description; meal_type, sodium_mg + basis; activity_type, intensity, steps; sleep_quality 1–5; stress_level 1–5 | source, updated_at |
| **medication_adherence_events (10)** | Each dose taken / missed / skipped | medication_id, status, scheduled_for, taken_at, note | source, reported_at |
| **journal_entries (8)** | Notes / questions for the doctor | category, entry_text | review status, reviewed_by / at (doctor) |
| **trusted_contacts (7)** | Emergency contacts | name, relationship, phone, consented_at, active | - |
| **emergency_events (12)** | Emergency screen history | trigger_reason, alert_id, symptoms_snapshot, started / ended_at, actions_attempted ("tapped", never "called"), reported_outcome, location (only with consent) | - |

### Doctor-authored

| Table | Purpose | Doctor fills | Notes |
|---|---|---|---|
| **care_plans (17)** | Targets and monitoring plan | target_systolic / diastolic, readings per day, instructions, lifestyle and follow-up text, effective_from / to, review_date, status, supersedes_id | One active plan per patient; default target 140/90 if none |
| **medications (12)** | Prescriptions and own meds | name, dose_text, schedule_text, active, started_on, ended_on | PULSE never computes or changes a dose |
| **doctor_notes (8)** | Private note + advice | note, advice, follow_up_at | Patients get only advice and follow-up date, through the API |

### Alerts and ML outputs (system)

| Table | Purpose | Fields | Who can act |
|---|---|---|---|
| **alerts (20)** | Rule results | alert_class, rule_code, rule_version, reason, actions, persistent, measurement_id, symptom_report_id, trigger_source, status + timestamps, resolution_note | patient acknowledges; doctor reviews, resolves |
| **alert_measurements (4)** | Readings of one alert episode | alert_id, measurement_id, patient_id, attached_at | read only |

| | | | |
|---|---|---|---|
| **ml_predictions (16)** | Next-day BP estimate | target_date, target, status (ok / insufficient_history), predicted_systolic / diastolic, 4 interval bounds, model_version, trained_on, features_until, n_readings_used | doctor reads only |
| **wearable_flags (17)** | Advisory ring flags | metric (+ multivariate, sleep), kind (outside_reference / personal_anomaly / multivariate_anomaly / early_warning, M9), window_kind (night / live_10min / forecast), window_start / end, value, score, reference_basis, band_id, model_version, review_status (open / confirmed / dismissed), reviewed_by / at, review_note | doctor updates review fields only |
| **patient_bands (14)** | Personal ring normal range | metric, statistic (night_median / night_min / live_10min_median), low, high, basis (reference / model_proposal / doctor), status (proposed → active → superseded, or rejected), proposed_model_version, approved_by / at, effective_from, supersedes_id | doctor approves via `private.approve_band` or rejects |
| **weekly_reports (14)** | Weekly summary per patient | week_start (local Monday), home_bp_category, control_status, next_week_above_target_prob, model_versions, summary (numbers only, ≤ 16 KB), status ready / reviewed, reviewed_by / at, doctor_comment | doctor reviews and comments |

## 4.4 Rules inside the database

| Rule | What it does |
|---|---|
| Validation `pulse-v1` (trigger, every reading) | **invalid** (kept, not used): systolic ≤ diastolic, systolic outside 50–300, diastolic outside 25–200, pulse outside 25–250, more than 5 minutes in the future. **needs_review**: systolic outside 70–250 or diastolic outside 40–150, an estimate (not a cuff), the same values within 2 minutes, a device-reported problem. **valid**: everything else. A very high reading (186/118) is valid data and an alert matter, not a data problem. |
| Checks (constraints) | Units fixed per metric; plausible ranges; ML predictions have values only if `ok`; a band's low < high, and an active band needs an approver; a flag's review time is set exactly when it's reviewed; report weeks start on Monday; summaries stay small and numeric. |
| `private.approve_band(band, doctor, low, high)` | One transaction: the active band for that metric becomes `superseded` and the proposed one becomes `active`. If the doctor edits low / high, the basis becomes `doctor`. Checks that the doctor is actively assigned and is the caller. It lives in the private schema, so it is not a public endpoint. |
| Audit triggers | Inserts and updates on alerts, care plans, bands, flags, reports and other sensitive tables are copied to `audit_log`. |
| One active row | One active care plan per patient; one active band per (patient, metric, statistic); one flag per (patient, metric, kind, window start); one rule firing per reading / symptom report. |

## 4.5 Who can see what (RLS + grants)

| Data | Patient | Assigned doctor | Admin | Logged out |
|---|---|---|---|---|
| Own profile, readings, context, devices | read + insert own | read | read | nothing |
| Care plan, medications | read | read + write | read | nothing |
| Doctor notes | no direct read (API shows advice only) | read + write | read | nothing |
| Alerts | read, acknowledge | read, review, resolve | read | nothing |
| ML outputs (predictions, flags, bands, reports) | **nothing** (decision M4) | read; review fields only | read | nothing |
| Ring samples | read own | read | read | nothing |

| | | | | |
|---|---|---|---|---|
| Audit log | - | - | read | nothing |

A doctor who is not assigned sees nothing about a patient. ML jobs and the seed use the privileged database connection; no app user can write ML outputs. The live REST API refuses logged-out requests (error 42501, tested).

# 5. Data: synthetic, demo and real

| Dataset | Kind | Size | Used for |
|---|---|---|---|
| `pulse-sim-v1` demo seed | synthetic | 12 patients × 30 days, 730 readings; ring 28 days | Cloud demo accounts; kept byte-stable by fingerprint tests |
| `pulse-sim-v2` ML cohort | synthetic | 200 patients × 90 days, 36,890 readings; 1,005,144 ring samples; 67 labelled ring episodes | Training and testing `mbp-v1.1`, `ring-v1.1`, `wcls-v1.1`, `mbp-v2.1` |
| NHANES 2017–March 2020 | real | 7,296 adults with pulse bands; 2,309 treated adults for `wrisk-v1` | Population pulse / BP bands; real classifier test |
| Mendeley pjcv33hpp7 | real | 1 patient, 128 home readings over 384 days | Calibrating the simulator's day-to-day variation |
| PhysioNet Apnea-ECG | real | 8 overnight SpO2 recordings | Testing the ring SpO2 rule on real nights |

## 5.1 How the BP simulator works (`pulse_data/simulate.py`)

Each synthetic patient has a baseline systolic (cohort: uniform 110–165 mmHg), a diastolic of about 0.62 × systolic, a slow trend, a chance of skipping a session, and (60 % of them) a medication with a chance of missing doses. Every day the generator draws the day's context, then a hidden **BP effect**:

**effect(day)** = latent drift + day noise + trend × day + **6** if a dose was missed yesterday + **1.2** × (sodium − 3,200 mg) / 1,000 − **2** if activity ≥ 30 min + **2** if sleep < 6 h
**latent drift** = φ × yesterday's drift + Normal(0, σ) · **reading**: systolic = baseline + effect + 4 (morning surge) + Normal(0, 5); diastolic = baseline + 0.55 × effect + 2 + Normal(0, 3.5), at least 15 below systolic; pulse ≈ Normal(72, 6), +3 in the morning.
Sessions around 07:00 and 20:00 (−40 to +60 min), 1–2 readings each; 0.3 % of readings have systolic and diastolic swapped on purpose, to test data cleaning.

The effects of missed doses, salt, activity and sleep are **invented** for testing the pipeline; they are not clinical estimates.

## 5.2 Calibrated on real home BP: v1 vs v2

| Systolic, 1 reading every 3 days | SD (mmHg) | change between readings | lag-1 correlation | Noise setting |
|---|---|---|---|---|
| **Real** patient (Mendeley, n = 128) | 11.7 | 9.97 | 0.40 | - |
| `pulse-sim-v1` (first generator) | 6.7 | 6.45 | 0.24 | φ 0.70, σ 3.0, no day noise |
| `pulse-sim-v2` (median of 111 patients) | 11.8 | 10.04 | 0.41 | φ 0.95, σ 3.6, day noise SD 6.0 |

The first generator was about half as variable as real home BP, which made the first model look too good (MAE 5.2). v2 was chosen from a small grid to match the real series; all models were retrained (`*.1`). Diastolic v2 6.8 / 5.8 / 0.41 vs real 6.4 / 5.1 / 0.37. One real patient gives a realistic size, not a population fit. The demo seed stays v1, so the cloud accounts did not change.

## 5.3 How the ring simulator works (values from published studies)

| Simulated value | Based on |
|---|---|
| Night resting HR ~ 65 ± 7 bpm between people, ~3 bpm night to night; every 10 min at night, hourly by day | Quer et al., PLOS ONE 2020: 92,457 wearable users, mean 65 bpm |

| | |
|---|---|
| Night SpO2 ~ 96.5 %, 6 samples per night; one 3–5 % dip on half the nights | Gries & Brooks, Chest 1996: lowest sleep saturation 90.4 % in healthy people. This is why PULSE uses the night **median** |
| COPD: personal SpO2 4 points lower | BTS 2017: target 88–92 % if at risk; MENA COPD 5.2 % men / 1.8 % women (BREATHE) |
| Sleep apnea: 2–4 dips of 4–9 % per night, night HR +3 | Desaturations ≥ 4 % linked to resistant hypertension (cohort of 284). The 30 % rate is an assumption |
| Type 2 diabetes: HRV (RMSSD) 0.92 SD lower | HRV in T2DM meta-analysis, PLoS ONE 2018 |
| Atrial fibrillation: noisy HR, meaningless HRV | Irregular pulse on wrist PPG (Apple Heart Study 2019); rates are assumptions |
| HRV ~ 34 ± 9 ms, skin temperature ~ 34 °C | assumptions (personal baseline only) |

**Invented coupling** (for testing only): night HR +0.25 bpm and HRV −0.3 ms per mmHg of the hidden BP effect. **Injected episodes** in the ML cohort (30 % of patients, 1–3 days, after day 35): `hr_rise` +15 bpm (18), `spo2_dip` −5 % (22), `hrv_drop` −40 % (27). 10 % of days have no ring data (ring not worn). Cohort conditions: sleep apnea 56, diabetes 32, COPD 8, atrial fibrillation 5 of 200. Ring values use a separate random generator, so the BP data never changes (tested).

## 5.4 The demo patients (fictional; risk factors from real Algerian statistics)

| Patient | Story | Sex, age | Smoking | Condition codes | BMI | Waist |
|---|---|---|---|---|---|---|
| Amina | controlled | F, 58 | never | hypertension | 26.5 | 86 |
| Karim | rising trend, gaps; seeded night HR rise | M, 65 | current | hypertension, diabetes, dyslipidemia, sleep apnea | 31.2 | 104 |
| Nadia | severe event + chest pain | F, 69 | never | hypertension, kidney disease | 33.0 | 99 |
| Yacine | sparse, **not assigned** | M, 47 | former | hypertension | 24.0 | 90 |
| Fatima | cohort; COPD story | F, 47 | never | hypertension, COPD | 24.8 | 73 |
| Khadija | cohort | F, 31 | never | hypertension | 26.9 | 89 |
| Samir | cohort | M, 43 | current | hypertension, diabetes, dyslipidemia, sleep apnea | 22.7 | 99 |
| Rachid | cohort | M, 44 | never | hypertension | 19.5 | 91 |
| Mourad | cohort | M, 66 | never | hypertension, dyslipidemia | 19.1 | 96 |
| Djamel | cohort | M, 42 | never | hypertension, dyslipidemia | 24.2 | 85 |
| Houria | cohort; atrial fibrillation story | F, 67 | never | hypertension, dyslipidemia, atrial fibrillation | 26.5 | 88 |
| Hocine | cohort | M, 55 | never | hypertension, diabetes | 17.9 | 85 |

Plus **Dr. Leila H.** (doctor, assigned to 11 patients) and an admin. The 8 cohort patients' sex, age band, smoking, diabetes, cholesterol, BMI band and abdominal obesity are drawn from the Algerian STEPS 2016–2017 shares (Moussouni et al. 2022): 55 % women among hypertensives; age 30–44 / 45–59 / 60–69 = 39.3 / 27.7 / 11.6 %; smoking 11.1 % current; diabetes 17.5 %; high cholesterol 29.6 %; obese 24.9 %; abdominal obesity 67.8 % (waist > 94 cm men, > 80 cm women). Height, salt answers, language and ring values are labelled assumptions. With 8 people the drawn shares can't match the published ones exactly.

## 5.5 Do the fields cover the Algerian blood-pressure factors?

| Factor (STEPS 2016–2017) | Where it lives | Status |
|---|---|---|
| Age (strongest: odds ratio 12.3 for 60–69 vs 18–29) | `patient_profiles.birth_year` | LIVE |
| Sex (34.2 % women vs 28.3 % men) | `patient_profiles.sex` | LIVE |
| Obesity, abdominal obesity (OR 1.7 each) | `body_measurements` : weight, height, waist; BMI computed | LIVE |

| | | |
|---|---|---|
| Cholesterol (OR 2.0), diabetes, kidney, heart, stroke, chronic illness | `condition_codes` (13 codes after M8) | LIVE |
| Salt / bread | `bread_portions_per_day`, `adds_salt_at_table`, `salty_processed_food`, meal `sodium_mg` | LIVE |
| Smoking, activity, sleep, stress | `smoking_status`; `lifestyle_logs` | LIVE |
| Ramadan fasting | `lifestyle_logs` category `fasting` (medication timing stays the doctor's decision) | LIVE |
| Language, local time | `preferred_language` fr / ar; timezone Africa/Algiers | LIVE |
| Emergency number | backend config: **14** (Protection civile), decided 30 Oct; never dialled in tests or the demo | DECIDED |

### 5.6 Real public datasets (official sources only, no Kaggle)

| Source | What it is | Terms | What we used | Limits |
|---|---|---|---|---|
| NHANES 2017–March 2020 (CDC / NCHS) | Real exams: 3 oscillometric BP and pulse readings (Omron HEM-907XL, 5 min seated), BMI, waist, smoking, diagnoses, survey weights | NCHS agreement: statistics only, no re-identification, no linkage | Weighted pulse / BP percentiles by sex × age; `wrisk-v1` | US, not Algeria; exam BP, not home BP; self-reported diagnoses; one visit |
| Mendeley pjcv33hpp7 (Dvornik et al. 2017) | 128 home readings (mean of 3 trials) of 1 treated 67-year-old man, every 3 days for 384 days | CC BY-NC 3.0: non-commercial, credit | Calibrating `pulse-sim-v2` | 1 patient; no clock times; drop it if PULSE becomes commercial |
| PhysioNet Apnea-ECG 1.0.0 (Penzel et al. 2000) | 8 real overnight SpO2 recordings: 4 apnea, 1 borderline, 3 control | ODC Attribution 1.0 | Testing the ring SpO2 rule | 8 nights; sleep-lab oximeter, not a ring |

**How they're handled:** `python -m pulse_ml.real_data fetch all` downloads into git-ignored `data/external/<source>/` and writes a `MANIFEST.json` (URL, licence, date, sha256). A changed file is never silently overwritten. Only aggregated numbers and trained models are committed, labelled `real:<source>`. Kaggle was rejected: it needs an account token, and the popular BP sets have unclear licences or origins.

### 5.7 Data rules that hold everywhere

- Every row says where it came from: `source` = manual / device_sync / csv_import / **simulated**; ML artefacts say `synthetic:pulse-sim-v2` or `real:nhanes`.
- Only **valid** readings feed averages, trends and ML; needs_review and invalid readings are counted, not used.
- No health values in logs or error messages; jobs print counts only.
- Raw real data never leaves the laptop; the demo password and the database URL live only in `.env`.

## 6. AI models: training, test data, results

| Model | Question it answers | Algorithm | Trained on | Writes | Status |
|---|---|---|---|---|---|
| `mbp-v1.1` | What will tomorrow's morning BP be? | ridge regression | synthetic v2 | ml_predictions | SERVED |
| `ring-v1.1` | Is tonight's ring data unusual for this person? | robust z + IsolationForest + references | synthetic ring | wearable_flags | IN USE (flags) |
| `wcls-v1.1` | Will next week's home mean be above target? | logistic regression | synthetic v2 | weekly_reports | IN USE (report) |
| `mbp-v2.1` | Does ring data improve next-day BP? | ridge regression | synthetic v2 + ring | - | EXPERIMENT |
| `wrisk-v1` | Is a treated adult's BP uncontrolled (≥ 140/90)? | logistic regression | **real** NHANES | - | NOT SHOWN |

All models are Python (pandas, numpy, scikit-learn 1.x, joblib). Each one is saved as `ml/artifacts/<version>/model.joblib` plus a `metadata.json` holding features, split, metrics and limits. Every number in this section is copied from those files. **None of them creates an alert, decides an emergency, or touches medication.**

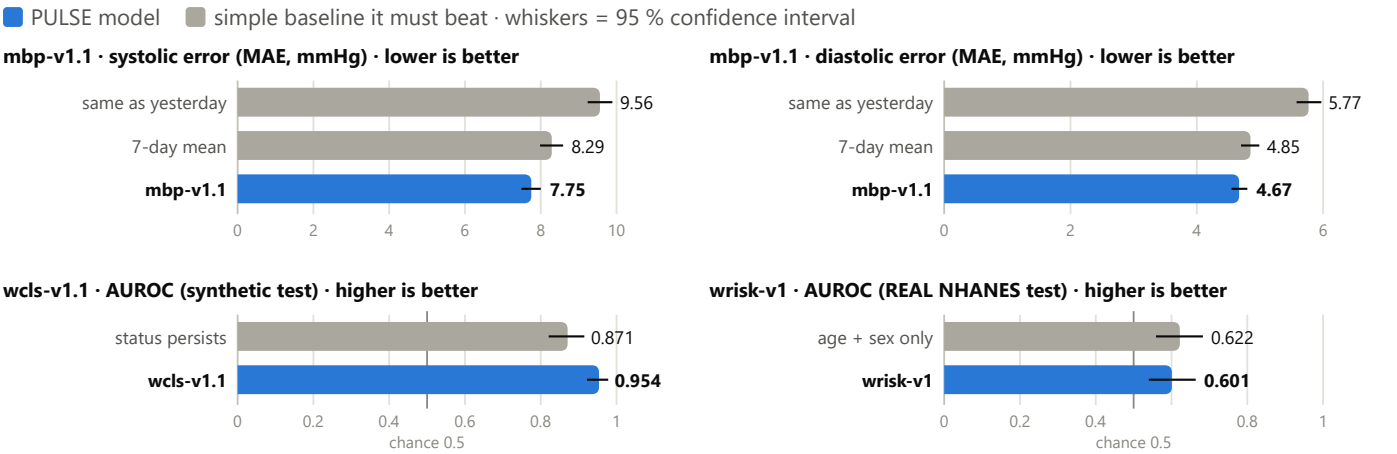



Figure 3. Each model against the simple method it has to beat, on held-out test data. mbp-v1.1 and wcls-v1.1 beat their baselines with confidence intervals that don't overlap. On real people, wrisk-v1 is no better than age and sex, so it isn't shown.

## 6.1 How every model is trained and tested (same recipe)

**1. Data**
Generate the labelled synthetic cohort, or load real data from `data/external/`

**2. Features**
Same code in training and serving (`features.py`, `ring.night_table`, `week_examples`)

**3. Split by patient**
70 / 15 / 15, seed 7; no patient in two sets; features only from earlier days

**4. Candidates**
A simple model (ridge or logistic) vs gradient boosting

**5. Choose on validation**
Pick the best on validation patients only

**6. Test once**
On unseen test patients, vs simple baselines; 95 % bootstrap CI over patients

**7. Save**
`model.joblib` + `metadata.json`, versioned folder

**8. Serve**
A batch job loads it and writes only its own table

Tests check the recipe: no patient overlap, no feature from the target day or later, deterministic retraining, and jobs that write only their own table.

## 6.2 `mbp-v1.1`: next-day morning blood pressure SERVED

| | |
|---|---|
| **Use** | Informational estimate in the **doctor's** view, labelled "Model estimate (synthetic training data)". Never shown to patients, never used for alerts or medication. |
| **Target** | Mean systolic and diastolic of the next patient-local day's readings before 12:00. |
| **Inputs (21 features)** | From valid readings of the 7 days before only: last morning value, 7-day and 3-day means, 7-day morning mean, 7-day slopes, systolic spread, morning−evening gap, reading count, days with readings, day of week (7 flags). |
| **Eligibility** | ≥ 6 valid readings on ≥ 4 local days in the 7 days before, and ≥ 1 morning reading; otherwise `insufficient_history` (no numbers). |
| **Training data** | `pulse-sim-v2`, 200 patients × 90 days (36,890 readings, seed 7). |
| **Split** | By patient: 140 / 30 / 30 patients → 8,730 / 1,918 / 1,929 patient-days. |
| **Model choice** | Validation MAE (mean of sys and dia): **ridge 6.18** vs gradient boosting 6.30 → ridge. |
| **Uncertainty** | Split-conformal interval from validation patients: ±16.2 systolic / ±9.6 diastolic; on test it covered the true value **90.1 % / 89.5 %** of the time (target 90 %). |

| Test: 30 unseen patients, 1,929 days | systolic MAE (95 % CI) | systolic RMSE | diastolic MAE (95 % CI) | diastolic RMSE |
|---|---|---|---|---|
| "same as yesterday's morning" | 9.56 (9.24–9.89) | 12.04 | 5.77 (5.58–5.97) | 7.31 |
| "7-day mean" | 8.29 (7.98–8.59) | 10.42 | 4.85 (4.70–4.99) | 6.11 |

| | | | | |
|---|---|---|---|---|
| **mbp-v1.1** | **7.75 (7.49–8.00)** | **9.80** | **4.67 (4.55–4.80)** | **5.87** |

**Reading:** about 7.8 mmHg off on systolic, better than both simple methods. The error is larger than the first version's 5.2 because the training data is now as variable as real home BP; this is the more honest number. **Real-data check:** the only open real home series has 1 reading every 3 days, so the model can't run on it; it was used to calibrate the simulator instead.

## 6.3 `ring-v1.1`: ring monitoring FLAGS ONLY

| | |
|---|---|
| **Use** | Advisory flags for the doctor, never an alert, never the safety path. |
| **Night statistic** | Median of each metric 00:00–06:00 local (HR, HRV, SpO2, skin temperature), built from raw `wearable_samples` by the same code in training and serving. |
| **Flag 1: reference** | SpO2 night median < 94 % (BTS; **88 % with COPD**) or resting HR > 100 bpm (Mayo). Live 10-minute windows need ≥ 2 samples. |
| **Flag 2: personal** | Doctor-approved band if one exists; else robust z = (value − median) / (1.4826 × MAD) over the 28 nights before (≥ 14 nights, confirmed anomalies left out), flag if \|z\| > **3.5**. |
| **Flag 3: multivariate** | IsolationForest on the 4 personal z-scores (contamination 0.01, a synthetic setting). Skipped with atrial fibrillation; HRV is never flagged with AF. |
| **Training data** | Synthetic ring data for the same 200 × 90 cohort (1,005,144 samples), 67 injected episodes. Split by patient 140 / 30 / 30, seed 7. |

| Method (synthetic test, 30 patients) | episodes found | flagged nights | night precision | false flags / patient-week |
|---|---|---|---|---|
| reference only | 2 / 7 | 140 | 0.029 | 0.39 |
| **personal baseline, z > 3.5** | **3 / 7** | 104 | 0.077 | 0.27 |
| IsolationForest | 1 / 7 | 23 | 0.130 | 0.06 |
| personal or IsolationForest | 3 / 7 | 107 | 0.075 | 0.28 |

| z limit (validation only) | episodes found | false / patient-week |
|---|---|---|
| 2.5 | 9 / 10 | 0.91 |
| 3.0 | 7 / 10 | 0.44 |
| **3.5 (kept)** | **7 / 10** | **0.22** |

**Why 3.5:** same recall as 3.0 with half the false flags. 2.5 would mean about one false flag per patient every week. The doctor's band makes the limit patient-specific. 7 and 10 episodes are small samples.

### Real-data test: PhysioNet Apnea-ECG (8 real nights)

| Recording | Group | hours | SpO2 median | minimum | % time < 90 | ODI-4 / h | ring-cadence median (6 samples) | Ring rule flags? |
|---|---|---|---|---|---|---|---|---|
| a01r | apnea | 8.2 | 91 | 66 | 45.3 | 70.5 | 82.0 | **yes** |
| a02r | apnea | 8.8 | 86 | 61 | 61.0 | 56.5 | 83.5 | **yes** |
| a03r | apnea | 8.7 | 92 | 50 | 26.1 | 57.7 | 87.5 | **yes** |
| a04r | apnea | 7.5 | 86 | 50 | 61.1 | 59.4 | 83.5 | **yes** |
| b01r | borderline | 8.0 | 96 | 51 | 0.5 | 1.5 | 96.0 | no |
| c01r | control | 7.8 | 96 | 57.5 | 0.1 | 0.3 | 96.0 | no |
| c02r | control | 8.1 | 95 | 62.5 | 0.3 | 2.1 | 95.0 | no |
| c03r | control | 7.6 | 95 | 69.5 | 0.1 | 0.4 | 96.0 | no |

**Reading:** the night-median rule flagged **4 / 4** apnea nights and **0 / 4** borderline / control nights. A night-**minimum** rule would fail, because even controls hit 57–69 % from sensor artefacts. This is a sleep-lab oximeter and only 8 nights: a sanity check, not a validation. The simulator's sleep apnea is milder than these severe cases (real ~60 dips per hour).

## 6.4 `wcls-v1.1`: will next week be above target? IN THE WEEKLY REPORT

| | |
|---|---|
| **Label** | Next week's mean of valid home readings ≥ the patient's target (systolic OR diastolic). Synthetic targets 135/85 or 130/80; default 135/85. |
| **Inputs (11 features)** | This week's mean systolic and diastolic, reading count, morning and evening systolic, slope, spread, last-3 systolic, gaps to target (sys, dia), "above target now". |
| **Split** | By patient, seed 7: 1,791 / 384 / 387 patient-weeks. |
| **Model choice** | Validation AUROC: **logistic regression, BP only 0.959**; gradient boosting BP only 0.941; logistic + ring 0.956; boosting + ring 0.944. Ring features didn't help, so BP only. |

| Test (387 weeks, 240 above target) | AUROC (95 % CI) | balanced accuracy | Brier (lower = better) | Confusion [TN FP / FN TP] |
|---|---|---|---|---|
| **wcls-v1.1** | **0.954 (0.922–0.978)** | 0.892 | **0.080** | 130 17 / 24 216 |
| "status persists" | 0.871 (0.821–0.915) | 0.871 | 0.124 | 125 22 / 26 214 |
| only the 48 weeks where status changes | 0.126 (0.054–0.197) | 0.210 | 0.508 | 5 17 / 21 5 |

**Reading:** better ranked and better calibrated than "nothing changes", but it has no skill on the weeks that actually change. It mostly says "same as now". So the report shows it as **"how close to the target line"**, not as a predictor of change.

## 6.5 `mbp-v2.1`: does ring data help predict BP? EXPERIMENT

Same split and recipe as `mbp-v1.1`, plus 7 ring features (last night's HR, HRV, SpO2, temperature; 7-night HR and HRV; ring nights in 7). Validation ridge 6.125 vs 6.178.

| Test MAE (synthetic) | systolic | diastolic |
|---|---|---|
| v1.1 features | 7.75 (7.49–8.00) | 4.67 (4.55–4.80) |
| + ring night features | 7.71 (7.45–7.97) | 4.64 (4.51–4.77) |

**Decision:** the 0.04 mmHg gain is within the noise, and the ring-to-BP link in the simulator is invented, so any gain is built in. `mbp-v1.1` stays served.

## 6.6 `wrisk-v1`: trained on REAL people (NHANES) NOT SHOWN

| | |
|---|---|
| **Question** | Is a treated adult's exam BP uncontrolled (mean ≥ 140/90)? |
| **Population** | US adults 18+ taking prescribed BP medicine, ≥ 2 valid exam readings: 2,309 people. Survey weights used as sample weights. |
| **Features (12, only fields PULSE stores)** | age, female, BMI, waist, current / former smoker, diabetes, high cholesterol, kidney disease, COPD, heart failure, stroke. |
| **Split** | By person, seed 7: 1,616 / 346 / 347. Weighted uncontrolled rate in test: 30.9 %. |
| **Model choice** | Weighted validation AUROC: logistic 0.568 vs gradient boosting 0.511 → logistic. |

| Test (347 people) | AUROC (95 % CI) | weighted | bal. acc. |
|---|---|---|---|
| wrisk-v1 | 0.601 (0.540–0.664) | 0.623 | 0.528 |
| age + sex only | 0.622 (0.559–0.683) | 0.611 | 0.500 |

| Calibration (test quintiles) | predicted | observed |
|---|---|---|
| 1 (lowest) | 0.22 | 0.31 |
| 2 | 0.29 | 0.26 |
| 3 | 0.34 | 0.39 |
| 4 | 0.38 | 0.41 |
| 5 (highest) | 0.46 | 0.47 |

Odds ratios per unit: female 1.38, stroke 1.37, kidney disease 1.32, diabetes 1.29, current smoker 1.23, age 1.016 per year, high cholesterol 0.76, COPD 0.62. With the 130/80 label: weighted AUROC 0.624. **Reading:** on real data, these risk factors can't tell whose BP is controlled under treatment; control has to be **measured**, which is exactly what PULSE does. A near-random score would mislead a doctor, so it isn't displayed. Limits: US population, exam BP, self-reported diagnoses, one visit.

## 6.7 Real population "safe band" for the cuff pulse (NHANES)

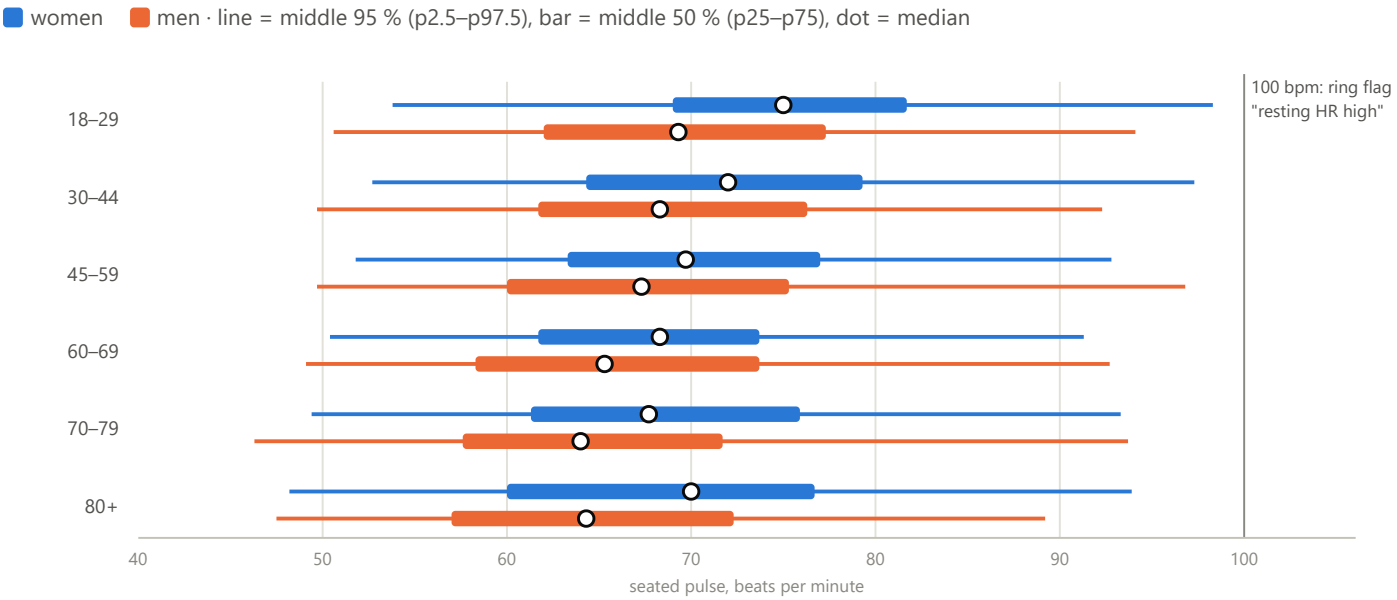



Figure 4. Real seated pulse of 7,296 US adults (NHANES 2017–2020, survey-weighted, mean of ≥ 2 valid readings). The middle 95 % is about 46–98 bpm in every group, so a pulse under 60 is common. That's why ring rules flag only a **high** resting HR.

**How PULSE uses it:** the weekly report places the patient's 7-day mean **seated cuff** pulse as a percentile among US adults of the same sex and age ( `summary.pulse_vs_population` ). It's context, not a threshold, and it is never applied to ring night HR (a different measurement). BP percentiles are also stored, but home BP categories use the ACC/AHA home column, because NHANES BP is exam BP.

## 6.8 Medical reference values (reviewed by the team's medical mentor, 30 Oct)

| Value | Used where | Source |
|---|---|---|
| Home BP categories: normal, elevated 120, stage 1 130/80, stage 2 **135/85** (= office 140/90) | weekly report `home_bp_category` (OR rule on the week's home mean) | ACC/AHA 2017, Table 11, home (HBPM) column |
| SpO2 < 94 % (COPD < 88 %) | ring flag (reference), band floor, R-RING-1 | BTS oxygen guideline 2017 |
| Resting HR > 100 bpm (high side only) | ring flag (reference), band ceiling | Mayo Clinic (adults at rest 60–100) |
| HRV, skin temperature | personal baseline only; HRV ignored with AF | no population reference used |
| Oximeter accuracy limits | wording, limits text | FDA safety communication, 19 Feb 2021 |

**Proposed safety rule R-RING-1, for the backend:** SpO2 median over 10 minutes < 94 % with ≥ 2 samples → prompt a cuff reading, a symptom check and re-seating the ring. Never an emergency by itself; symptoms keep their own emergency path. It runs on raw `wearable_samples` in the backend, so the safety path never waits for an ML job. Spec: `ml/pulse_ml/reference.py` ( `R_RING_1` ). Reviewed by the medical mentor on 30 Oct. **TO BUILD** (backend)

## 6.9 30-day day-by-day test on patients no model has seen (30 Oct)

**How it was run** ( `ml/pulse_ml/backtest.py` ):

- **New patients:** 100 synthetic patients × 90 days, made with another random seed and other IDs ( `H0001…` ; training used `S0001…` ). No patient, day or ring profile overlaps the training data, and a test checks this.
- **Frozen models:** no model was retrained or tuned on this data.
- **Day by day:** days 1–60 are history; the test days are 2–31 March. On each test day every model sees only data from before that day, through the same code as in serving, and its output is compared with what actually happened that day.

| Next-morning BP: 2,271 patient-days (65–82 patients a day; 11 too little history) | systolic MAE (95 % CI) | RMSE | diastolic MAE (95 % CI) | within 10 mmHg |
|---|---|---|---|---|
| "same as yesterday" | 9.50 (9.21–9.80) | 11.85 | 5.48 (5.30–5.65) | 60.9 % |
| "7-day mean" | 8.41 (8.17–8.69) | 10.51 | 4.83 (4.66–5.01) | 65.9 % |
| **mbp-v1.1** (served) | **7.88 (7.65–8.11)** | **9.80** | **4.56 (4.41–4.71)** | **68.6 %** |
| mbp-v2.1 (+ ring, experiment) | 7.88 (7.65–8.11) | 9.79 | 4.56 (4.41–4.71) | 68.9 % |

**Reading:**

- `mbp-v1.1` had a lower error than "same as yesterday" on **30 / 30** days and lower than the 7-day mean on **28 / 30** days.
- Its 90 % interval covered the real value **90.7 %** (systolic) and **91.0 %** (diastolic) of the time.
- The error on new patients (7.88) is close to the original test (7.75), so the model did not just memorise its training patients.
- Ring data adds nothing: `mbp-v2.1` beat v1.1 on only 15 / 30 days, a coin flip.

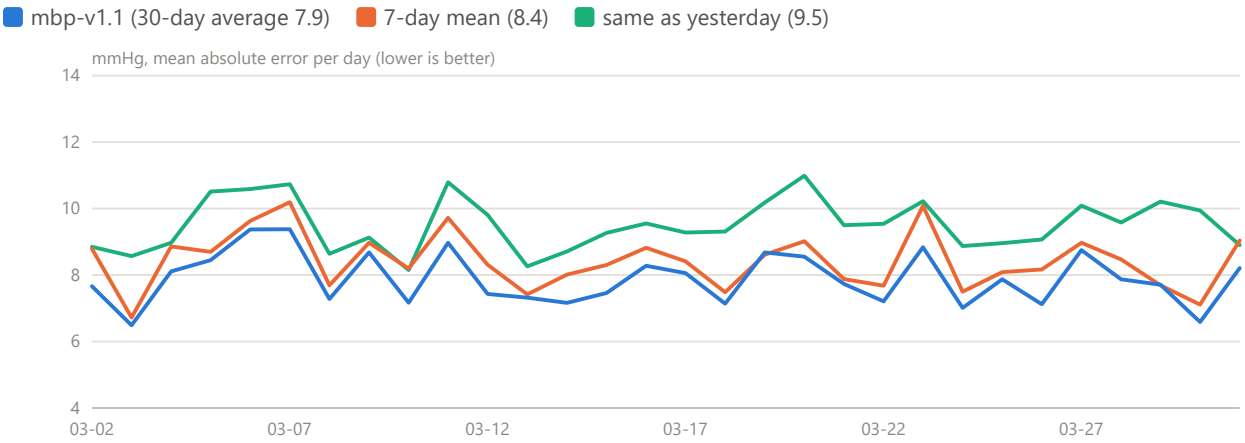



Figure 6. Systolic error on each of the 30 test days. The blue line (the model) is the lowest on almost every day.

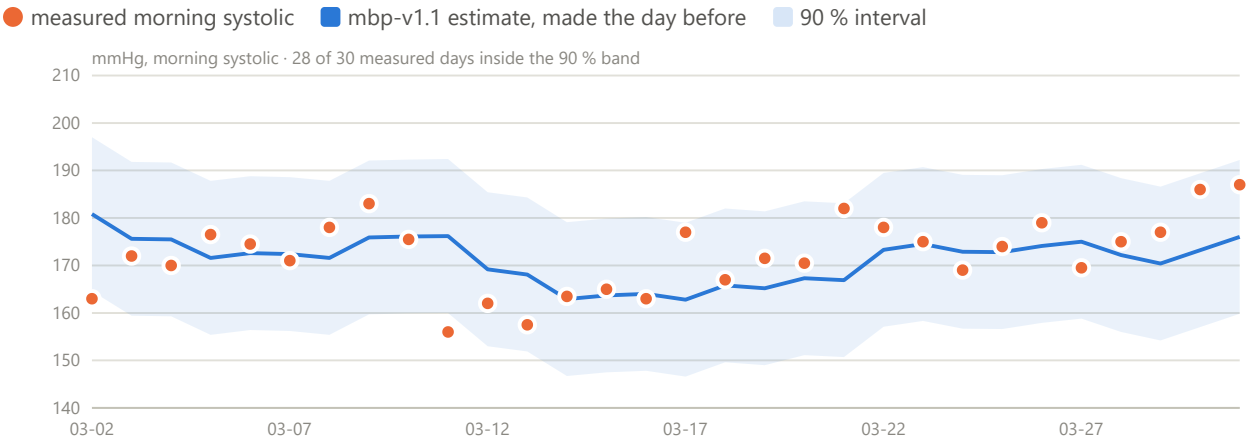



Figure 7. One unseen patient (H0020, synthetic) over the 30 test days. Each estimate was made the evening before, from earlier days only. 28 of the 30 measured mornings fall inside the band.

| Ring, every night: 2,703 nights of 100 patients, 14 planted events | events found | flagged nights | precision | other flags / patient-week |
|---|---|---|---|---|
| medical limit only (SpO2 < 94 %) | 6 / 14 | 443 | 0.02 | 1.12* |
| **personal baseline, z > 3.5** | **13 / 14** | 107 | 0.22 | **0.22** |
| IsolationForest | 9 / 14 | 44 | 0.32 | 0.08 |
| personal or IsolationForest | 13 / 14 | 117 | 0.21 | 0.24 |

The personal baseline found every heart-rate rise (7 / 7) and every HRV drop (2 / 2), and 4 / 5 oxygen dips. *433 of the 443 medical-limit flags come from the 31 sleep-apnea patients, whose night oxygen really is low: they're real findings, counted here only because they aren't planted events. 14 events is a small sample; on the original test the same rule found 3 / 7.

| Weekly classifier: 400 predictions (100 patients × 4 weeks, 285 above target) | AUROC (95 % CI) | balanced accuracy | Brier | Confusion [TN FP / FN TP] |
|---|---|---|---|---|
| **wcls-v1.1** | **0.963 (0.937–0.980)** | 0.873 | **0.072** | 95 20 / 23 262 |
| "status persists" | 0.881 (0.831–0.919) | 0.881 | 0.100 | - |
| only the 40 weeks where status changed | 0.055 | 0.076 | 0.555 | - |

The same pattern as the original test: well ranked and well calibrated, but no skill on the weeks that actually change. It stays shown as "how close to the target line".

| REAL people: wrisk-v1 on NHANES Aug 2021–Aug 2023 (a later survey it never saw; 1,851 treated adults, 31.1 % uncontrolled) | AUROC (95 % CI) | weighted | Brier | bal. acc. |
|---|---|---|---|---|
| wrisk-v1 (trained on 2017–2020) | 0.596 (0.568–0.624) | 0.592 | 0.210 | 0.508 |
| age + sex only (same training people) | 0.565 (0.534–0.597) | 0.555 | 0.213 | 0.500 |

On new real people it is stable (0.60 again) and slightly above age and sex, but still weak, so it stays hidden from reports. The survey renamed two variables in 2021–2023 (exam weight `WTMEC2YR`; BP medicine `BPQ150`); the meaning was checked against the CDC codebooks. **Not testable on real data:** next-day BP and the weekly classifier need daily home BP of many patients, and no open dataset has that. The ring SpO2 rule's real test (8 PhysioNet nights) is in 6.3.

## 6.10 Live ring: warn BEFORE a limit is crossed (`live-v1`, 30 Oct)

**What it does:** inside `monitor --follow`, every new minute of ring data is checked within about 44 ms (p95 81 ms) per patient. It uses everything up to that minute, including the reading that arrived a second ago.

- **Forecast:** HR, SpO2 and skin temperature in 10 and 20 minutes.
- **Early warning:** if the value is still inside its limit but the chance of crossing it within 20 minutes is high, the **doctor** gets a warning (M9 `early_warning` row). Limits: SpO2 94 % (COPD 88 %), resting HR 100 or the doctor's band, temperature normal + 1 °C.
- **Night going badly:** from 01:00, every 30 minutes, the chance that the night ends restless.

**Tested on:** 700 simulated training nights and 700 **unseen simulated** test nights. Patients see nothing, and the backend safety rules stay in charge.

**SIMULATOR-ONLY RESULTS.** Every figure in this section (warning rates, lead times, false warnings per night, restless-night figures, forecast errors, speed) comes from 700 + 700 simulated nights. None of it is real-world or clinical validation.

| Unseen simulated nights | nights | warned before the limit | warning came (median) | old "limit crossed now" rule |
|---|---|---|---|---|
| oxygen falling | 93 | **93.5 %** | **17 min before** | 7 min after |
| heart rate climbing | 103 | **97.1 %** | **20 min before** | 9 min after |
| fever starting | 73 | **94.5 %** | **17 min before** | (no rule) |

**Nights with nothing planted:**

- Early warnings per night: normal 0.04 (140 nights), loose ring / bathroom spike 0.03 (73), restless 0.11 (158), sleep apnea 0 (60).
- The low oxygen of sleep apnea is flagged by the old rule on 98 % of those nights.

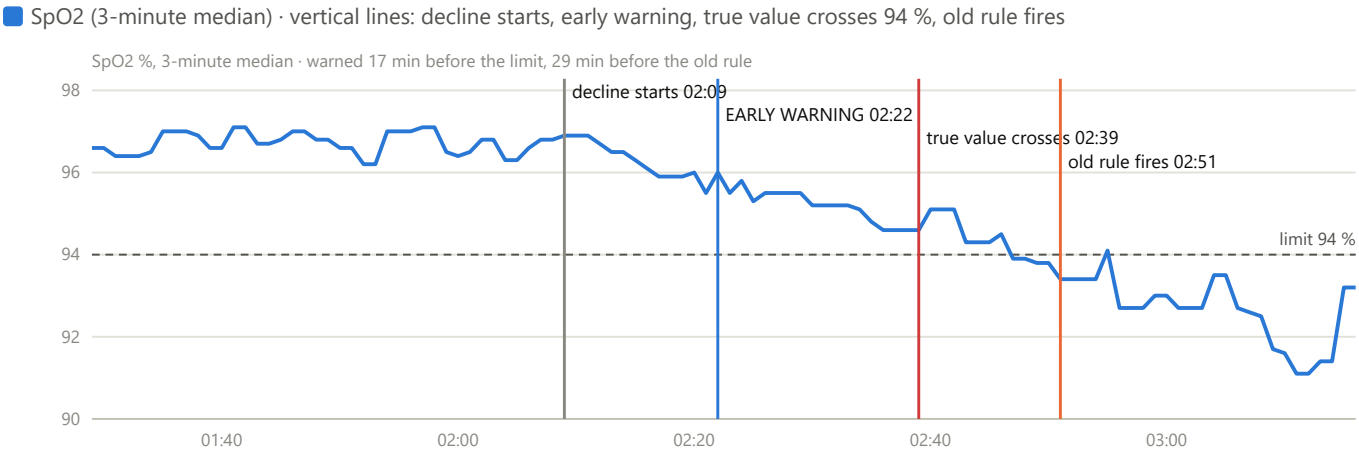



Figure 8. One unseen simulated night with falling oxygen, replayed minute by minute. The doctor gets the early warning at 02:22. The true oxygen level crosses 94 % at 02:39, and the old rule reacts only at 02:51.

| Next-minutes forecast, mean absolute error (unseen simulated nights) | +10 min | +20 min | "same as now" (+20) | straight-line trend (+20) |
|---|---|---|---|---|
| heart rate (bpm) | 2.56 | **2.86** | 3.61 | 7.43 |
| SpO2 (%) | 0.38 | **0.39** | 0.52 | 1.05 |
| skin temperature (°C) | 0.058 | **0.064** | 0.078 | 0.171 |

**Restless night:** 94.9 % of restless nights were warned, by 01:00 (median), and 1.5 % of other nights were warned. AUROC: 0.966 at 01:00, 0.986 at 03:00.

**How it was built:**

- A first linear forecast learned that blips come back to normal and never warned.
- Gradient-boosting models learn how a real decline differs from a passing dip, wake-up or loose ring.
- Thresholds were set on held-out training nights (no more than 0.05 false warnings per night): HR 0.35, SpO2 0.45, temperature 0.85.
- Replaying 20 nights one minute at a time gave exactly the same warnings.

**Limits:** the situations, slopes and artefacts come from my simulator. These numbers show the mechanism, not real-world accuracy, which needs a real ring and doctor-reviewed flags.

| **Clinician review: all of these were reviewed and accepted by the medical mentor on 30 Oct (prototype use). The results above stay simulator-only.** | |
|---|---|
| SpO2 limit as a forecast trigger | 94 %; COPD 88 % |
| Heart rate limit during sleep | 100 bpm (or the doctor's live band) |
| Skin temperature limit | personal night normal + 1.0 °C |
| P(limit crossed within 20 min) threshold | heart rate 0.35; SpO2 0.45; skin temperature 0.85 |
| Horizon · persistence · quiet period · freshness | 20 min · 2 minutes in a row · 30 min per metric · newest sample ≤ 2 min old |
| Restless night | P ≥ 0.7, from 01:00, every 30 min |
| `ring-v1.1` night personal anomaly | robust z > 3.5 against the last 28 days |
| R-RING-1 (backend safety rule) | SpO2 10-min median < 94 % with ≥ 2 samples → cuff reading, symptom check, re-seat the ring |

# 7. Live ring monitoring and the weekly doctor loop

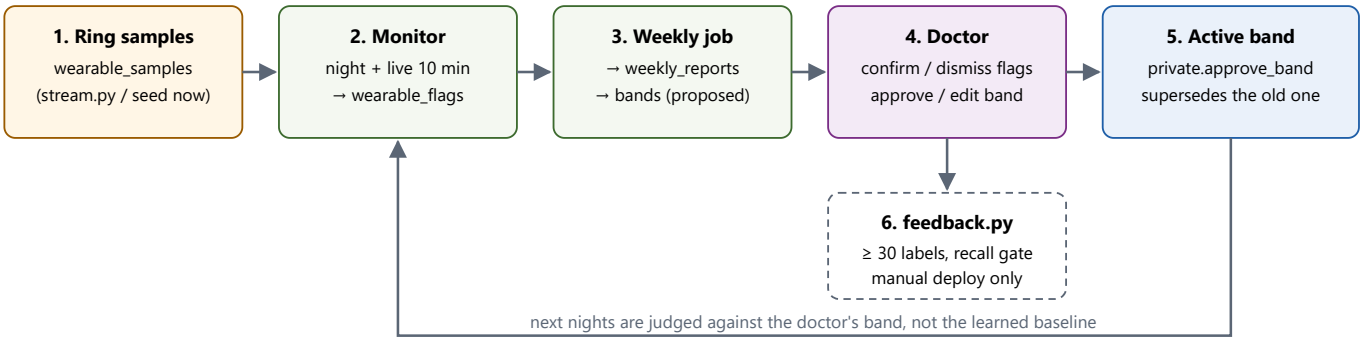



Figure 5. The weekly loop. The model only proposes; a doctor activates a band; labels can retune personal thresholds, never the medical references, and never deploy themselves.

| Piece | Rule |
|---|---|
| Monitor, night mode | `monitor --nights` : every assigned patient's nights; reference, then band or personal z, then IsolationForest. One flag per (patient, metric, kind, window), so re-runs never duplicate. |
| Monitor, live mode | `monitor --follow --every 5` : polls new samples by `created_at` with a 5-minute overlap (catches late rows); 10-minute windows; band flags need 2 windows in a row. |
| Weekly report | Patient-local Monday–Sunday. `home_bp_category` from valid home readings; `control_status` vs the active care-plan target (default 140/90; insufficient_data under 3 readings); `wcls-v1.1` probability when ≥ 3 valid readings; ring night summary, flag counts, latest BP estimate, pulse placement (NHANES). A report the doctor already reviewed is never overwritten. |
| Band proposal | Median of the last 28 nights ± max(3 × 1.4826 × MAD, floor). Floors: HR 5 bpm, HRV 5 ms, SpO2 1.5 %, temperature 0.5 °C. Clipped to the reference on the safety side (SpO2 low ≥ 94, or ≥ 88 with COPD; HR high ≤ 100); SpO2 capped at 100. Needs ≥ 14 nights; none for HRV with AF. |
| Feedback (review M5) | Doctor-reviewed flags become labels; ≥ 30 needed; they retune only `personal_anomaly` thresholds; a retrained model must not lose injected-episode recall ( `feedback --gate` ); no automatic deployment. |

## Verified on the local copy (30 Oct 2026, all 12 demo patients with a simulated ring)

- `monitor --nights` : 34 flags. Karim's seeded HR rise is caught on its 2 nights. Karim and Samir (sleep apnea) get low night SpO2 flags. Fatima (COPD, night SpO2 ~92 %) gets **no** false SpO2 flag thanks to the 88 % limit; before the COPD rule she had 5. Houria (AF) gets no HRV flag.
- Live: `stream --scenario spo2_dip` + `monitor --follow` gives Nadia a live SpO2 flag within about 10 seconds.

- `weekly` : 11 reports, 43 band proposals (e.g. Fatima's SpO2 band 89.8–92.8 %, waiting for the doctor). A test approves and edits Karim's HR band through RLS, re-runs the monitor, and checks that the flags now cite `band:<id>` .

## 8. How the jobs run (inference contract)

| Job | Reads | Writes | Rules |
|---|---|---|---|
| `predict` | Valid readings of actively assigned patients, strictly before the target day in the reading's own time zone | One row per patient, target day, model version in `ml_predictions` (upsert) | Default target = tomorrow (patient-local). Values and intervals null unless `ok` . `trained_on` = `synthetic:pulse-sim-v2` . Unassigned patients (Yacine) get no row. |
| `monitor` | `wearable_samples` , active bands, condition codes, confirmed flags | `wearable_flags` only | Advisory. Never alerts. |
| `weekly` | Readings, care plans, ring nights, flags, predictions, profile + body measurements | `weekly_reports` , proposed `patient_bands` | Numbers-only summary; reviewed reports are kept. |

**Backend mapping:** endpoint 24 reads the latest prediction for (patient, date), newest model first; no row → `model_unavailable` . Still to build by the backend: weekly report endpoint, flag review, band approval through `private.approve_band` , R-RING-1. All jobs print counts only, never health values.

## 9. Tests and quality

| File | tests | What it guards |
|---|---|---|
| `tests/test_db.py` | 49 | Validation policy cases; duplicates flagged not dropped; labelling constraints; import dedup and same-file refusal; one active care plan; patient / doctor / admin / logged-out visibility; forbidden writes; doctor notes hidden; ML tables doctor-only; Algerian fields; wearable checks; band approval supersedes and is audited; live REST API enforces access. |
| `tests/test_ingest.py` | 5 | CSV parsing, row-level rejections, missing columns, unknown time zone, deterministic seed labelled simulated. |
| `tests/test_simulate_stability.py` | 7 | SHA-256 fingerprints of the BP data (ring and condition changes can't alter BP); the 30-day holdout cohort is unseen; ring samples and live situations pass the table checks; the cloud refresh deletes only simulated demo-ring samples; chronic conditions shape ring values. |
| `ml/tests/test_live.py` | 6 | Falling oxygen and climbing heart rate are warned before the limit; loose ring and bathroom spike are not; whole night = minute by minute; limits follow COPD / AF / bands; 2-minutes-in-a-row and 30-minute quiet rule; the live monitor writes one early warning and nothing else (rolled back). |
| `ml/tests/test_ml.py` | 6 | Home-BP categories; no patient overlap; features only from earlier days; short history → insufficient; deterministic training; the predict job writes only predictions for assigned patients. |
| `ml/tests/test_ring.py` | 7 | CSV and database give identical nights; conditions change references; baseline uses earlier nights and skips confirmed anomalies; doctor band overrides baseline; live windows; late rows caught; seeded episode flagged. |
| `ml/tests/test_weekly.py` | 3 | Band maths; regression gate blocks lower recall; full weekly report and doctor band loop through RLS. |
| `ml/tests/test_classify.py` | 3 | Next-week label with the OR rule; features never see the label; deterministic training; the report gets a probability. |
| `ml/tests/test_real.py` | 9 | Raw real data is git-ignored; weighted percentiles; NHANES mapping and population; PULSE fields map to the same features; Mendeley parsing; pulse placement band; ODI-4 counts real desaturations. |

**Latest full run (30 Oct 2026): 95 / 95 passed** on the local database (the live-login tests run with the demo password loaded from `.env`). Every test write is rolled back: the demo data is unchanged.

## 10. Commands (Windows, repo `Desktop\pulse`)

| Command | What it does |
|---|---|
| `npx -y supabase@2.122.0 start` / `db reset --local` | Local database (ports 55321–55323), all migrations + seed |
| `.venv\Scripts\python -m pulse_data.simulate seed` | Write the demo seed (password from `.env`) |
| `.venv\Scripts\python -m pulse_data.simulate csv` | ML cohort to `data/synthetic/` (git-ignored) |
| `cd ml` then `..\.venv\Scripts\python -m pulse_ml.train` | Train + test `mbp-v1.1` (~70 s) |
| `... -m pulse_ml.ring_train` · `classify` · `mbp_v2` | Train + test `ring-v1.1`, `wcls-v1.1`, `mbp-v2.1` |
| `... -m pulse_ml.real_data fetch all` | Download the 3 real datasets (~200 MB) + manifests |
| `... -m pulse_ml.real_bands` · `wrisk` · `real_tests all` | NHANES bands, `wrisk-v1`, real-data tests |
| `... -m pulse_ml.predict --all` · `monitor --nights` · `weekly` | The three jobs (local databases only; any other host is refused, no unlock) |
| `... -m pulse_data.stream --scenario spo2_dip --minutes 1` | Simulated live ring (refuses any non-local database, no unlock) |
| `... -m pulse_data.live_sim --out data/live/test --seed 2027` · `... -m pulse_ml.live_test` | Minute-level situations; train + minute-by-minute test of `live-v1` |
| `... -m pulse_ml.monitor --follow` + `... -m pulse_data.stream --patient-email patient.b@demo.pulse.test --scenario oxygen_falling --speed 5 --minutes 20` (`DATABASE_URL` = local dev stack) | Live demo: an early warning before the limit (adds simulated samples to the local database) |
| `scripts\cloud_update.ps1` | Cloud: pending migrations + ring demo refresh + check (**DONE** 30 Oct) |
| `scripts\run_cloud_jobs.ps1` | Cloud: run the 3 jobs (`DATABASE_URL` from `.env`, or a temporary login from the linked Supabase CLI) (**DONE** 30 Oct; now fails closed, the jobs refuse any non-local database) |
| `.venv\Scripts\python -m pytest` | All tests (local stack running) |

## 11. Safety, ethics and limits

- **ML is advisory only.** No model creates alerts, decides an emergency, or suggests starting, stopping or changing medication. Symptoms trigger the emergency path on their own.
- **Patients never see ML output** until the backend owner and a clinician decide what a patient should see.
- **Synthetic is always labelled.** Metrics on synthetic data show the pipeline works; they say nothing about real patients.
- **Not validated for Algeria.** NHANES is the US population; Mendeley is one patient; Apnea-ECG is 8 lab nights.
- **Data terms respected:** NHANES statistics only (no re-identification, no linkage); Mendeley non-commercial; PhysioNet attribution.
- **Before any real use:** real, consented daily home readings; retraining and evaluation by patient and over time; clinician review of thresholds, wording and display; drift monitoring.

## 12. Next steps

| # | Step | Who | Status |
|---|---|---|---|
| 1 | Run the 3 ML jobs on the cloud to fill the 4 ML tables, then a read-only check | Data/AI | DONE 30 Oct, then frozen by Ayoub |
| 2 | Share the demo password privately; send the backend the message in `docs/data/CLOUD_STEPS.md` | You | DONE 30 Oct |
| 2b | M9 is in the cloud (kept by Ayoub, 31 Oct). Turning on `monitor --follow` / live-v1 against the cloud needs Ayoub to lift the freeze | Ayoub, then backend | FROZEN |
| 3 | Backend: alert rules writing to Supabase; weekly report, flag review (now including early warnings) and band approval endpoints; R-RING-1 | Backend | TO BUILD |
| 4 | Frontend: doctor panels for the BP estimate, ring flags, weekly report, band approval; profile fields | Frontend | TO BUILD |
| 5 | Clinician review: 180/120, red flags, SpO2 94 / 88 %, band rule, wording | Medical mentor | DONE 30 Oct |
| 5b | Clinician review of every live-v1 threshold and parameter, the ring-v1.1 z 3.5 and R-RING-1 (list in 6.10) | Medical mentor | DONE 30 Oct |
| 6 | Algerian emergency number: **14**. Backend sets `dz-14` to verified in `api/app/config.py` | Team, then backend | DECIDED 30 Oct |
| 7 | Choose a real device (validated upper-arm cuff recommended) and build its adapter | Team, then data/AI | MISSING |

## 13. Sources


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Alert thresholds, SpO2 limits, the band rule and wording were reviewed by the team's medical mentor on 30 Oct 2026 for prototype use; validation limits are engineering defaults. Study figures are quoted from the sources above.